Biochemistry Cume September 9, 2011

1. An enzyme of total mass 59,200 daltons, composed of two identical subunits, is assayed in a 10-mL reaction volume at constant temperature using varying initial substrate concentrations, shown in the table below. The amount of substrate converted to product by 1 ng of enzyme during the first 1 minute of reaction time is measured and recorded in the table below.

μM	nmol min^{-1}
[S]	v_o
1	0.11
3	0.25
5	0.34
10	0.45
30	0.58
50	0.61

   a. (30 points) Does the enzyme obey Michaelis-Menten kinetics? If so, give appropriate evidence, and determine the value of K_M.
   b. (30 points) What is the turnover number per enzyme active site under the given experimental conditions? (Assume two active sites per enzyme molecule.)

2. (40 points) A second unrelated enzyme has a mass of 120,000 daltons and consists of six identical subunits. For this enzyme, one unit of activity is defined as the amount of enzyme needed to hydrolyze 10 μmol of substrate in 15 min at 37 °C under standard conditions. The enzyme displays Michaelis-Menten kinetics. When the substrate concentration is equal to K_M, the purified enzyme (100% pure) exhibits 1400 units of activity per mg enzyme. What is the turnover number?
(State any assumptions that you make.)

Possibly useful information: $v_o = V_{max}([S])/([S] + K_M)$.

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